



PRIMARY SCIENCE

TOPICAL REVISION

ENVIRONMENT



SUMMARIZED SCIENCE REVISION NOTES



POLLUTION

To pollute is to contaminate, make dirty or impure.

SOIL POLLUTION

SOIL POLLUTANTS

1. Excess use of farm chemicals
 - Excessive use of fertilizers, pesticides end up killing useful organisms in the soil e.g. earth worms.
2. Acid rain
 - Sulphur dioxide in the air mixes with rain water to form acid rain.
 - Acid rain kills small animals in the soil
 - It also affects germination
3. Waste from industries
 - Chemical wastes make soil to support life
4. Floods carry waste and soluble chemicals from different sources. They end up polluting the soil.
5. Human waste – untreated human and domestic refuse could contain chemicals that affect micro-organisms.
6. Oil spillage – blocks air into the soil.
7. Plastic wastes – they are not decomposable (non-biodegradable)
8. Human activities e.g. mining, burning vegetation
 - Oil spillage interferes with air circulation in the soil.
 - Farm chemicals interfere with the soil structure and microorganisms.
 - Plastic affects root development.
 - Lower crop yields and quality
 - Destroys habitat for soil animals
 - Cause starvation to animals because plants are primary producers



SOIL CONSERVATION

This is conservation of the soil so that it can maintain fertility

It is conserved through

- Reduced use of excessive farm chemicals
- Recycling plastics instead of dumping them in the soil.
- Avoiding the burning of vegetation.
- Planting cover crops and other proper ways of land cultivation e.g. contour farming.
- Construction of gabions and terraces.
- Afforestation and re-afforestation
- Mulching
- Controlled grazing or proper stocking rate
- Avoid dumping of industrial waste on soil

AIR POLLUTION

This is the act of contaminating the air with toxic substances.

Ways in which the air is polluted

- a. Smoking cigarettes
Carbon dioxide, tar and nicotine cause respiratory diseases.
- b. Gases from vehicle exhausts.
Such gases contain harmful chemicals such as sulphur and lead
- c. Burning tyres and plastic materials
When they burn, they produce toxic gases
- d. Spraying farm chemicals
Some particles which are harmful to living things remain suspended in the air
- e. Aerosol sprays – they interfere with the ozone layer.
- f. Industrial waste gases e.g. sulphur dioxide

Effects of air pollution on plants

- Dust particles, soot and smoke block stomata.
- It affects photosynthesis
- It interferes with transpiration and gases exchange.



- Acid rain damages the leaves and roots of the plants.
- Acid rain interferes with living organisms in the soil.
- Decomposition of animals and plant materials is affected.

Effects of air pollution on animals

- It affects the respiratory system resulting in sneezing, coughing and bronchitis.
- Acid rain may kill aquatic animals
- Polluted air may lead to lung cancer.

Effects of air pollution on non-living things

Acid rains eat away (corrode) some metals e.g. iron sheets and limestone (marble)

Smoke and smog cause poor visibility, increasing accidents on the roads.

Dust contaminates our environment

Ways of controlling air pollution

- a. Not burning tyres and plastic materials that emit toxic substances.
- b. Vehicles should have well-maintained engines.
- c. Reducing the use of farm chemicals e.g. pesticides and use biological and mechanical methods of control.
- d. Avoid the use of harmful aerosol sprays which deplete the ozone layer. Use ozone -friendly chemicals.
- e. On-site treatment e.g. industrial gases
- f. Ban cigarette smoking in public.